

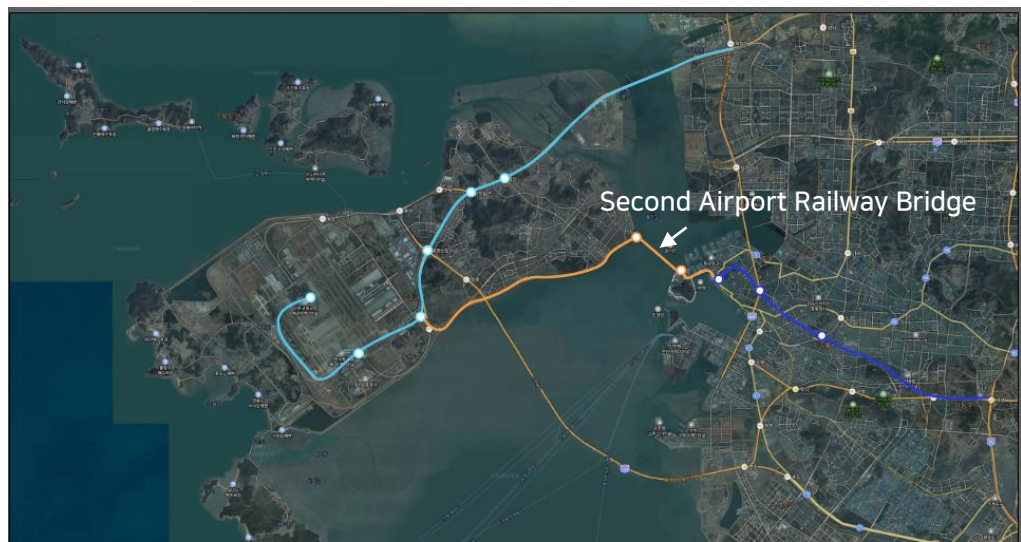


# Capstone Design in Civil and Environmental Engineering and Techopreneurship (1)

## Design of the Second Airport Railway Bridge Using PSC Box Girders

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### 1. Location

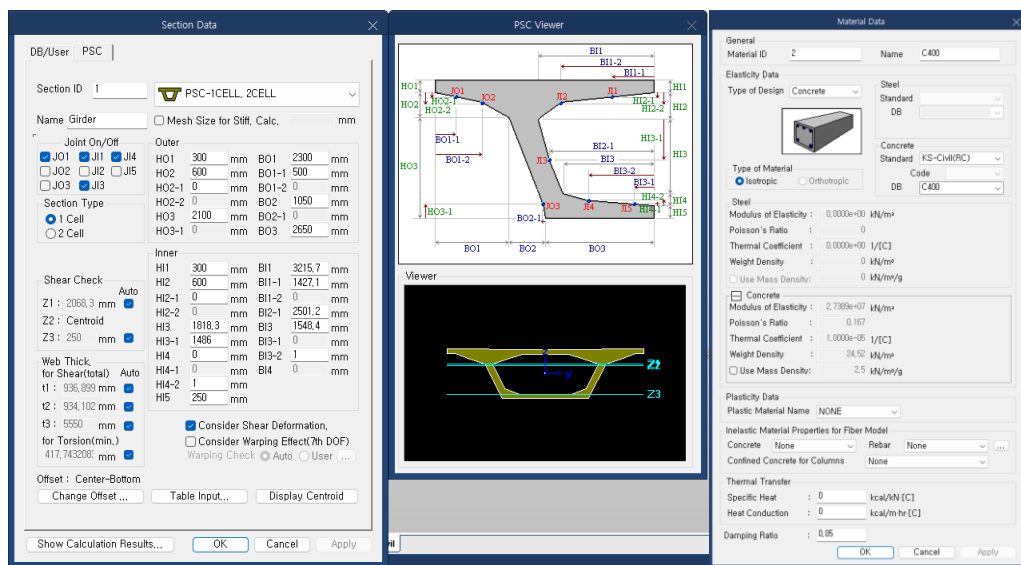


#### EXPECTATIONAL EFFECTS

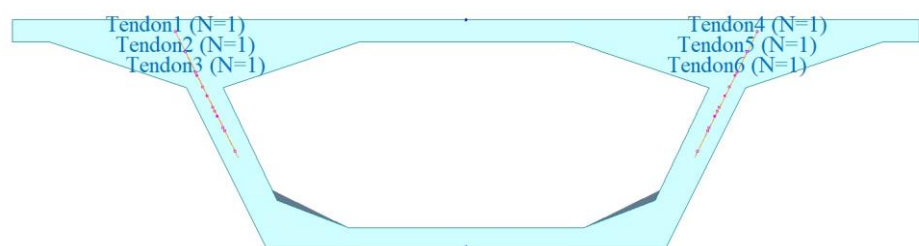
- Reduction of travel time between Wolmido and Yeongjongdo
- Improvement of accessibility from Yeongjongdo residential areas to Incheon and Seoul
- Enhancement of operational flexibility and route diversity for the Airport Railroad
- Expansion of tourism in Wolmido and Yeongjongdo

### 2. PSC Box Girder Bridge

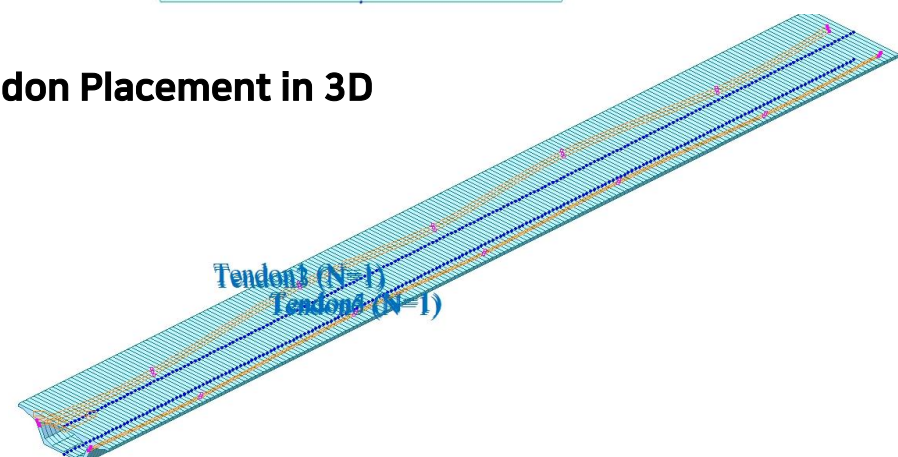
#### Section Dimension & Steel Properties



#### Tendon Placement



#### Tendon Placement in 3D



#### Initial Prestress

$$P_i = A_c \cdot f_{cci} = 9.49 \times 10^6 \times 4.62 = 43,843,800 \text{ N}$$

#### Tendon eccentric distance

$$e_{p1} = \frac{M_{d1}}{P_i} = \frac{-2.88 \times 10^{10}}{43,843,800} = -656.88 \text{ mm}$$
$$e_{p2} = \frac{M_{d1}}{P_i} = \frac{43,843,800}{7.6 \times 10^9} = 534.4 \text{ mm}$$
$$e_{p0} = \frac{M_{d1}}{P_i} = \frac{43,843,800}{43,843,800} = 170.21 \text{ mm}$$

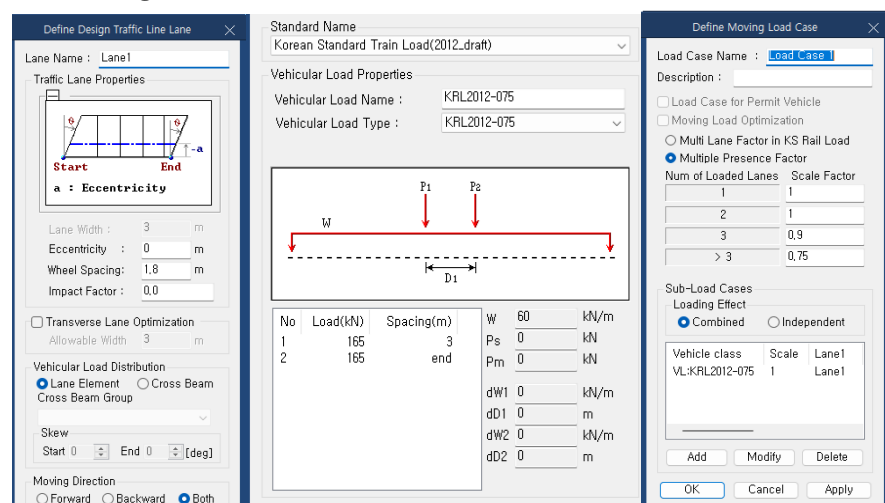
### 3. Design Load

- Self Weight : 232.13kN/m

- Non-structural Weight : 104.94kN/m

(TCL+ PCL + Rail + Railroad tie + Wires + Signal cables + Hand rail)

#### Moving Load

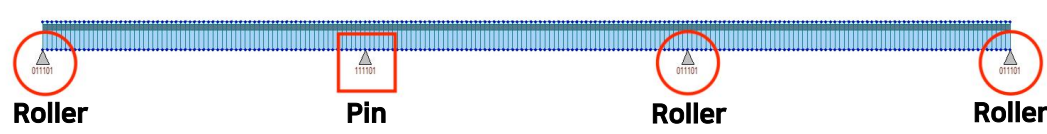


#### Load combination

| Load Case       | Factor |
|-----------------|--------|
| DL(ST)          | 1.0000 |
| PS(ST)          | 0.8000 |
| 2ndDL(ST)       | 0.0000 |
| Load Case 1(MV) | 1.0000 |

### 4. MIDAS Analysis

#### Boundary Condition

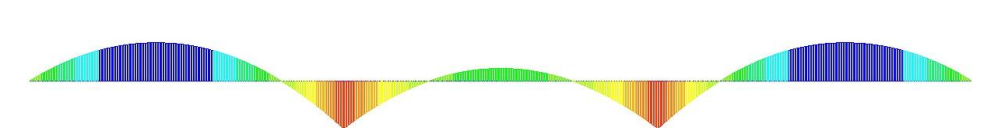


#### Moment Value



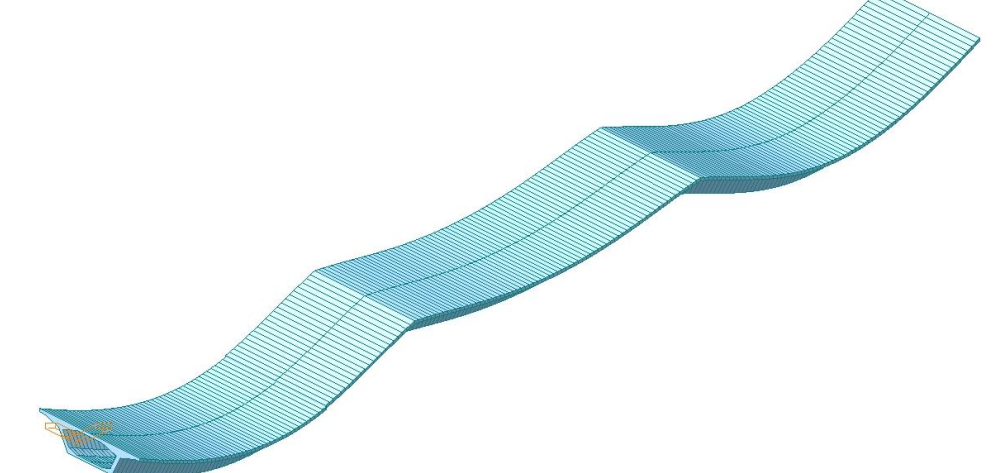
Max positive moment : 80,173.5 kN · m  
Max negative moment : -92,075.0 kN · m

#### Stress Value



| Unit(MPa)                   | Allowable Compressive Stress | Design Compressive Stress | Allowable Tensile Stress | Design Tensile Stress |
|-----------------------------|------------------------------|---------------------------|--------------------------|-----------------------|
| Immediately after Prestress | 16.46                        | 6.40                      | -1.56                    | 4.61                  |
| Service Load                | 17.64                        | 12.50                     | -3.13                    | -2.84                 |

#### Deflection



Max deflection: 17.08 mm  
Allowable deflection: L/800=43.75mm

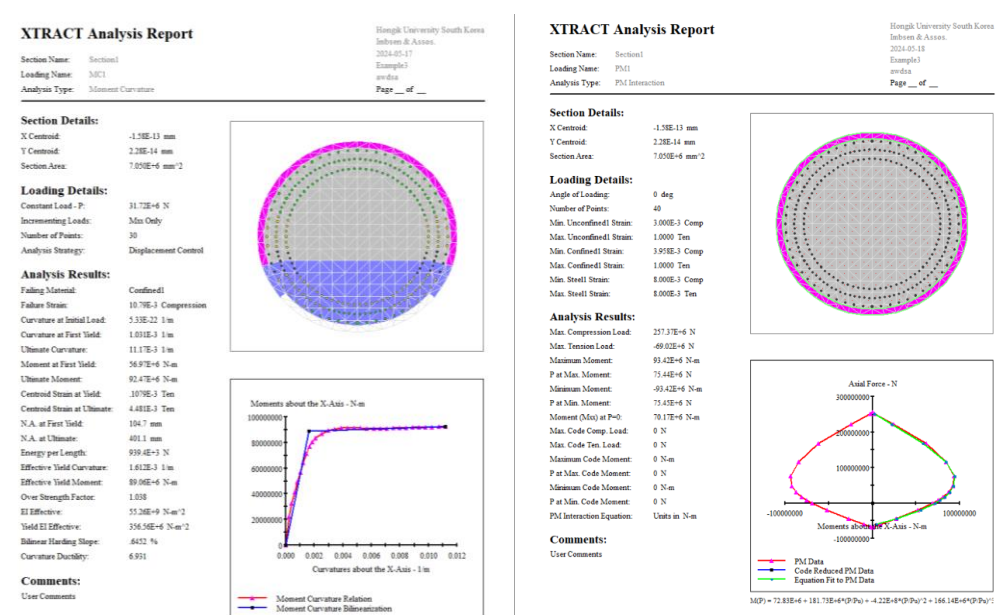
#### Reaction



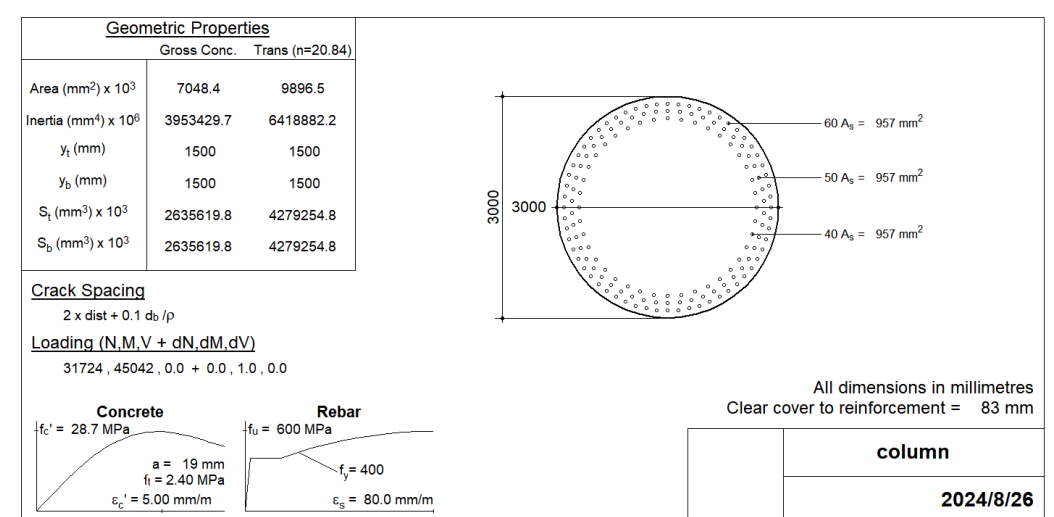
Reaction = 31,001.7 kN

### 5. Column Design and Information

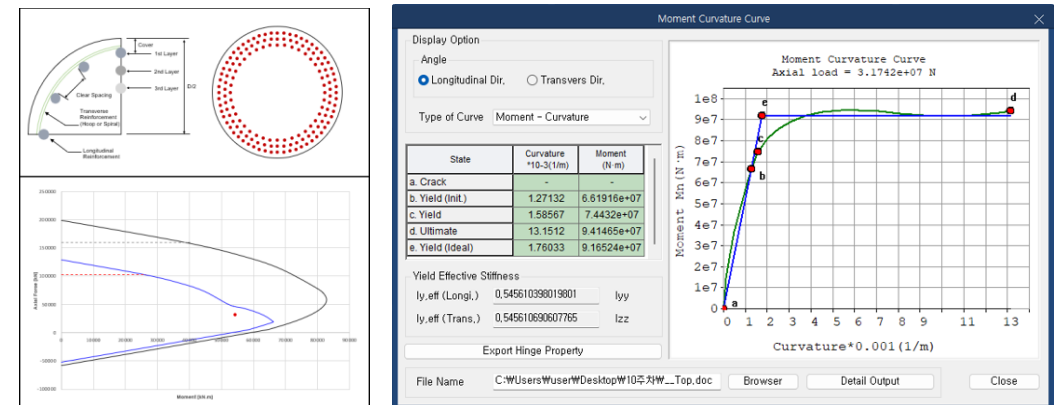
#### Section Information (Xtract)



#### Section Information (Response2000)



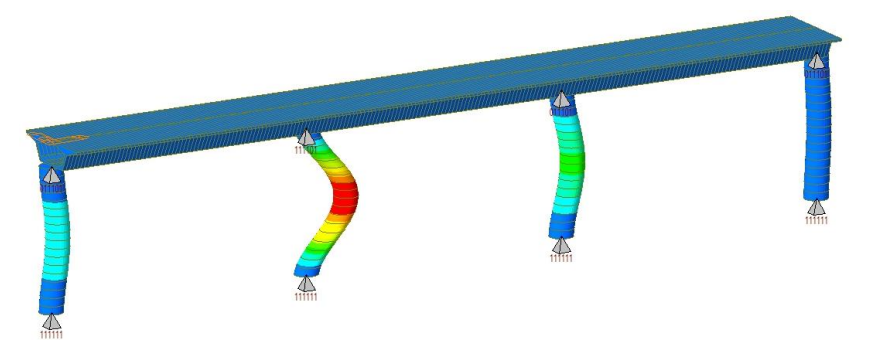
#### Section Information (MIDAS)



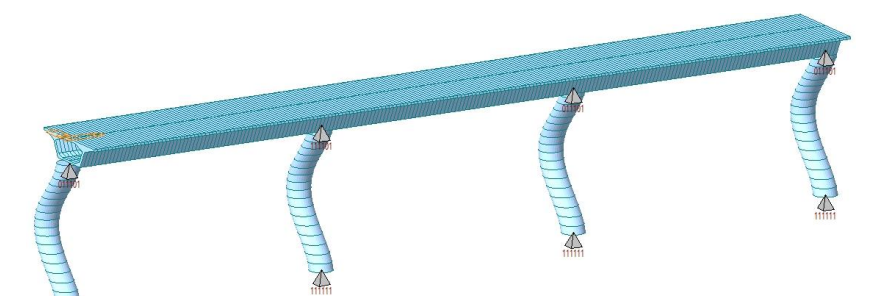
### 6. Natural Periods of Structure

#### Periods (Bridge Stiffness Analysis)

$$P_L = 1,067.5 \text{ kN} \quad m_L = 108,817.5 \text{ Kg}$$
$$P_v = 355.83 \text{ kN} \quad m_v = 36,272.2 \text{ Kg}$$
$$K = 1,283,336.9 \text{ N/mm}$$
$$T_L = 0.057873 \text{ s} \quad T_v = 0.033404 \text{ s}$$



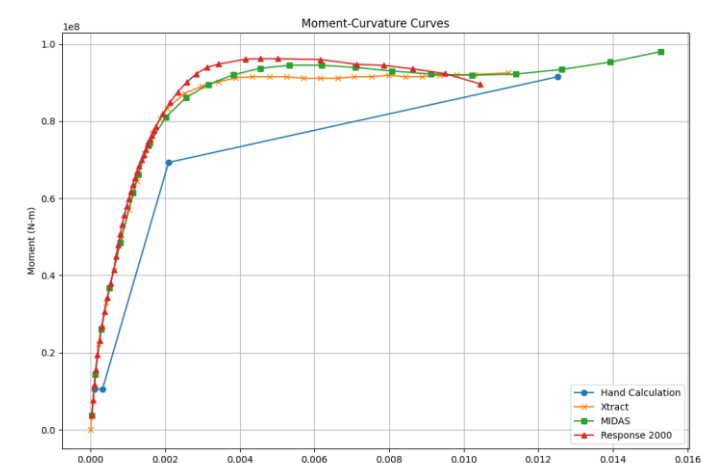
$$\omega_{xL} = \sqrt{\frac{1,283,336.9}{108,817.5}} = 108.5978 / \text{s} \quad T_L = 0.036388 \text{ s}$$



$$\omega_{xv} = \sqrt{\frac{1,283,336.9}{36,272.2}} = 188.0977 / \text{s} \quad T_v = 0.030846 \text{ s}$$

### 7. Moment Curvature & P-M Curve

#### Moment Curvature



$$t = 2,400 \text{ mm}, d' = 400 \text{ mm}, d = 2,000 \text{ mm}, d - d' = 1,600 \text{ mm}$$

$$E_c = 24,761 \text{ MPa}, E_s = 2.0 \times 10^5 \text{ MPa}, n = \frac{E_s}{E_c} = 8.63$$

<Uncracked Section>

$$I = 4.11 \times 10^{12} \text{ mm}^4, M_{cr} = \frac{f_r I_g}{y_t} = 1.06 \times 10^{10} \text{ N} \cdot \text{mm}$$

$$\phi_{cr} = \frac{M_{cr}}{E_c I} = 1.11 \times 10^{-7} \text{ mm}^{-1}$$

<Cracked Section>

$$I_{cr} = 1.4303 \times 10^{12} \text{ mm}^4, \phi_{cr} = \frac{M_{cr}}{E_c I_{cr}} = 3.19 \times 10^{-7} \text{ mm}^{-1}$$

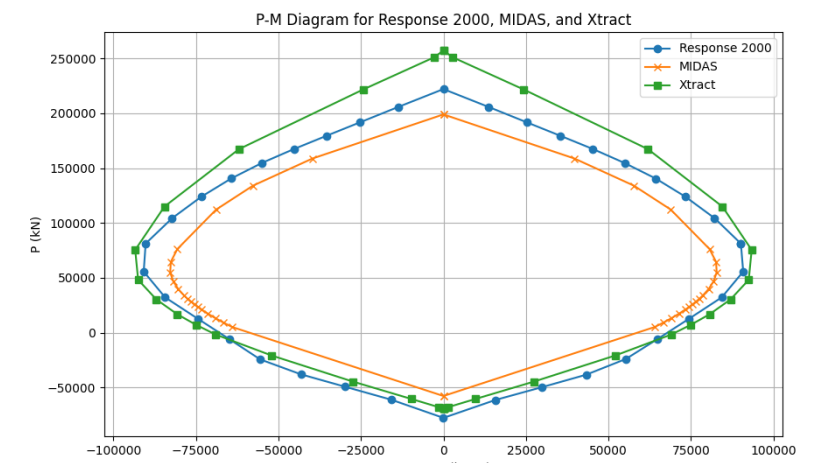
<Yielding Point>

$$M_y = \phi_y (EI)_{cr} = 6.93 \times 10^{10} \text{ N} \cdot \text{mm}$$
$$\phi_y = 2.09 \times 10^{-6} \text{ mm}^{-1}$$

<Nominal Point>

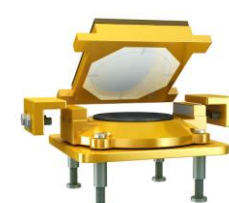
$$M_n = C_c (d - \beta c) + C_s (d - d') = 9.15 \times 10^{10} \text{ N} \cdot \text{mm}$$
$$\phi_n = 1.25 \times 10^{-5} \text{ mm}^{-1}$$

#### P-M Curve



### 8. Bridge Other Facilities

#### Bridge Bearing : Spherical Bearing



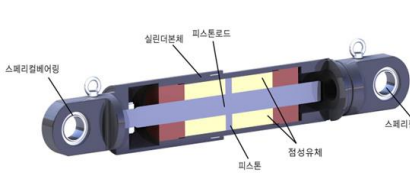
- Load transfer to the ground through a hemispherical bearing plate constrained by the upper and lower plates

#### Expansion Equipment :New Monocell joint



- Butt-type structure
- Wave structure on the vehicle contact surface
- Waterproofing and Excellent workability

#### Seismic Damper : LUD & STU



- Load reduction and displacement suppression
- Reduction of construction costs
- No maintenance required